



SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)



SIMATS ENGINEERING

COURSE NAME: TRANSMISSION LINES AND WAVE GUIDES

COURSE CODE: ECA15

COURSE OUTCOME COVERED

CO3 : Evaluate and design matching networks using quarter-wave transformers, single and double stub methods, and Smith chart-based analysis to optimize power transfer and minimize reflections in real-world RF and communication systems.

Assessment Tool Number : CO3_AT1

Assessment Tool Weightage : 40%

Assessment Tool : Multi-Tool Design Task

Duration : 30 minutes

Marks : 50 marks

Type of Assessment conduction : Hybrid

Evaluation Type : Manual Correction

Topics Covered: Impedance matching: Quarter wave transformer, Impedance matching by stubs, Single stub and double stub matching, Smith chart, Solutions of problems using Smith chart, Single and double stub matching using Smith chart.

Description:

- Apply theoretical concepts and use appropriate engineering tools to solve the given problem.
- Ensure correct design methodology, calculations/simulations, and technical accuracy.
- Analyze the results and clearly justify the design decisions using engineering principles.
- Work in hybrid mode using multiple tools and upload the developed code/files to

GitHub.

- Provide a demonstration in the next day's class, with assessment based on creativity, documentation, presentation, and timely completion.

Q. No.	Assessment Question	BTL Level
1	Define impedance matching and explain its importance in RF and microwave communication systems.	BTL 1 – Remember
2	Describe the working principle of a quarter-wave transformer for impedance matching.	BTL 2 – Understand
3	Explain the construction and significance of the Smith Chart in transmission line analysis.	BTL 2 – Understand
4	Compare quarter-wave transformer, single-stub, and double-stub matching techniques based on their operating principles and applications.	BTL 2 – Understand
5	Calculate the characteristic impedance of a quarter-wave transformer required to match a 50 Ω transmission line to a 100 Ω load.	BTL 3 – Apply
6	Determine the location and length of a single shunt stub required to match a given complex load using the Smith Chart.	BTL 3 – Apply
7	Design a double-stub matching network for a specified load impedance and verify the matching condition using CST Studio.	BTL 3 – Apply
8	Perform Smith Chart analysis to determine the reflection coefficient, VSWR, and input impedance of a mismatched transmission line.	BTL 3 – Apply

9	Simulate a quarter-wave transformer in an HFSS Simulation Software and analyze the S11 parameter at the design frequency.	BTL 3 – Apply
10	Demonstrate the impedance transformation along a transmission line using Smith Chart and validate the results through HFSS Simulation software.	BTL 3 – Apply
11	Analyze the effect of frequency variation on the performance of a quarter-wave transformer using simulated S-parameters.	BTL 4 – Analyze
12	Investigate how stub position influences impedance matching and reflection coefficient in a single-stub matching network.	BTL 4 – Analyze
13	Compare simulated and theoretical values of return loss, VSWR, and reflection coefficient for a double-stub matching network.	BTL 4 – Analyze
14	Analyze the impedance trajectory on the Smith Chart for different transmission line lengths and interpret the results.	BTL 4 – Analyze
15	Evaluate the effectiveness of quarter-wave transformer and stub matching methods for broadband and narrowband applications.	BTL 5 – Evaluate
16	Assess the impact of fabrication tolerances on the performance of impedance matching networks using an EM simulation tool .	BTL 5 – Evaluate
17	Evaluate the suitability of single-stub and double-stub matching techniques for waveguide applications with practical constraints.	BTL 5 – Evaluate
18	Design an impedance matching network for a practical RF communication system using the Smith Chart and verify its performance through an EM simulation tool.	BTL 6 – Create
19	Develop an optimized matching network that minimizes return loss over a specified frequency band using HFSS simulation tools.	BTL 6 – Create

20	Design and simulate a transmission line matching network for a microwave amplifier or antenna, compare different matching techniques, and justify the selected design based on S-parameter performance and power transfer efficiency.	BTL 6 – Create

_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of disciplinary action.

Signature of the student

RUBRICS FOR CALCULATION:

Criteria	Weightage (Marks)	Excellent (L4)	Good (L3)	Satisfactory (L2)	Needs Improvement (L1)
Problem Identification	20	Clearly identifies and defines the problem and requirements.	Identifies most requirements correctly.	Partial understanding of the problem.	Poor understanding of the problem.
Design Planning	20	Develops a comprehensive and logical design plan.	The design plan is mostly complete and logical.	The design plan is partially complete.	The design plan is inadequate.
Technical Accuracy	25	Design is accurate and meets all specifications.	Minor technical errors present.	Several technical errors present.	Major errors affecting functionality.
Creativity & Innovation	15	Demonstrates exceptional creativity and originality.	Shows good creativity.	Limited creativity.	No evidence of creativity.
Documentation & Presentation	20	Excellent organization, clarity, and presentation.	Good presentation with minor issues.	Adequate presentation.	Poor presentation and documentation.

Single Shunt Stub Matching Using the Smith Chart

Aim

To match a complex load impedance to the characteristic impedance of a transmission line using a single shunt stub (short-circuited or open-circuited) connected in parallel with the transmission line.

Theory

A single shunt stub is a section of transmission line connected in parallel with the main transmission line at a suitable distance from the load. The stub may be either short-circuited or open-circuited at its end. Its purpose is to cancel the reactive part of the load admittance so that the overall input admittance becomes equal to the characteristic admittance of the transmission line.

Since the stub is connected in shunt, it contributes only susceptance (B) to the circuit. Therefore, the impedance of the load must first be converted into admittance before the matching process begins.

The Smith Chart greatly simplifies the design by allowing the engineer to determine both the location of the stub and its length graphically without lengthy calculations.

The matching process consists of moving from the load point toward the generator until the normalized conductance becomes unity (1). At this point, only the susceptance remains to be cancelled. A stub having an equal and opposite susceptance is then connected at that location, resulting in a perfect impedance match.

Procedure

Step 1: Normalize the Load Impedance

Normalize the load impedance with respect to the characteristic impedance .

where

- = Load impedance
- = Characteristic impedance of the transmission line
- = Normalized load impedance

Normalization allows the impedance to be plotted directly on the Smith Chart.

Step 2: Convert Impedance into Admittance

Since the stub is connected in parallel, convert the normalized impedance into normalized admittance.

where

- = Normalized load admittance

Locate this admittance point on the admittance Smith Chart (or rotate the impedance point by 180° on the Smith Chart to obtain the corresponding admittance point).

Step 3: Draw the Constant VSWR Circle

Draw the constant VSWR circle passing through the normalized admittance point.

Every point on this circle represents the same reflection coefficient magnitude but a different position along the transmission line.

Step 4: Move Toward the Generator

From the load admittance point, move clockwise (toward the generator) along the constant VSWR circle until the normalized admittance becomes

or

At these points,

- Normalized conductance
- Normalized susceptance

The movement from the load to either of these points represents the distance between the load and the stub position.

Usually two such points exist on the Smith Chart, giving two possible matching solutions.

Step 5: Measure the Stub Location

Read the wavelength scale marked "Wavelengths Toward Generator (WTG)".

The difference between the reading at the load and the reading at the selected point gives

expressed in wavelengths (λ).

Step 6: Determine the Required Stub Susceptance

Read the susceptance value at the selected point.

Suppose the admittance is

Then the stub must provide

Similarly, if

then

The objective is

so that

which corresponds to a perfectly matched transmission line.

Step 7: Determine the Stub Length

The length of the stub depends on whether it is

- Short-circuited
- Open-circuited

(a) Short-Circuited Stub

Locate the short-circuit point on the Smith Chart.

Move along the constant-conductance circle until the required susceptance

is reached.

The distance moved from the short-circuit point gives the

(b) Open-Circuited Stub

Locate the open-circuit point on the Smith Chart.

Move along the chart until the required susceptance is obtained.

The distance travelled gives the required open-stub length.

Important Observations

- A shunt stub supplies only susceptance.
- The conductance must first be adjusted to unity before connecting the stub.
- There are generally two possible stub locations, each with a corresponding stub length.
- Either solution provides perfect impedance matching.
- The stub length is usually less than 0.5.
- Both short-circuited and open-circuited stubs can be used, although short-circuited stubs are generally preferred because they have lower radiation losses and are easier to fabricate.

Result

The complex load impedance is matched to the transmission line using a single shunt stub.

The following parameters are obtained from the Smith Chart:

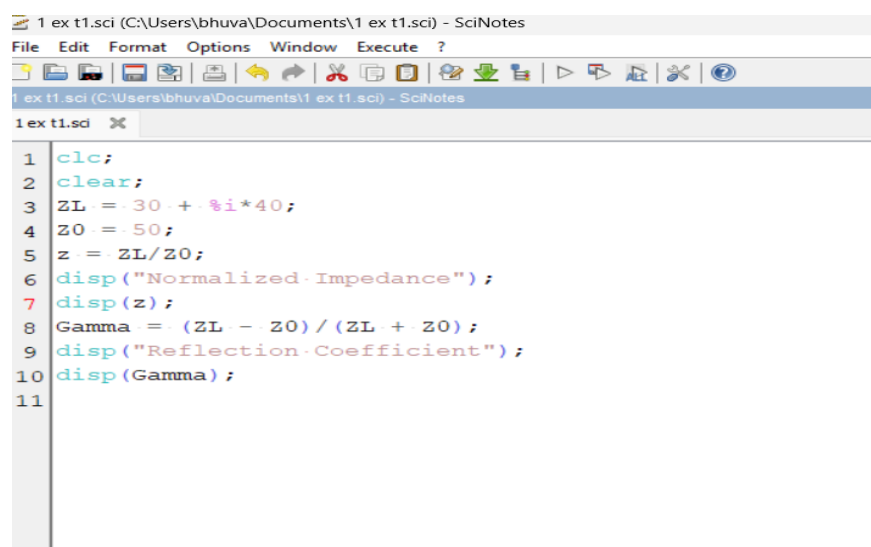
- Stub Location: Distance from the load to the point where the normalized admittance becomes 1 .
- Required Stub Susceptance: Equal and opposite to the susceptance at the stub location.
- Stub Length: Length of the short-circuited or open-circuited stub required to provide the cancelling susceptance.

After connecting the stub at the calculated location, the input admittance becomes

which corresponds to a matched transmission line with:

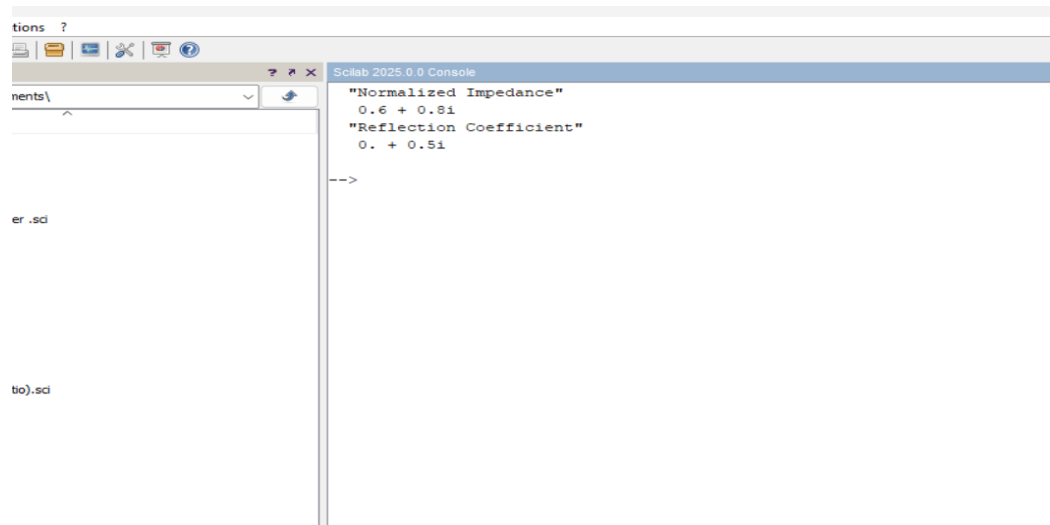
- Reflection coefficient,
- $VSWR = 1$
- Maximum power transfer from the source to the load
- No reflected power on the transmission line.

Scilab



```
1 ex t1.sci (C:\Users\bhuva\Documents\1 ex t1.sci) - SciNotes
File Edit Format Options Window Execute ?
1 ex t1.sci (C:\Users\bhuva\Documents\1 ex t1.sci) - SciNotes
1 ex t1.sci X
1 clc;
2 clear;
3 ZL = 30 + %i*40;
4 Z0 = 50;
5 z = ZL/Z0;
6 disp("Normalized Impedance");
7 disp(z);
8 Gamma = (ZL - Z0) / (ZL + Z0);
9 disp("Reflection Coefficient");
10 disp(Gamma);
11
```

OUTPUT:



```
Scilab 2025.0.0 Console
"Normalized Impedance"
0.6 + 0.8i
"Reflection Coefficient"
0. + 0.5i
-->
```

Explanation of Each Line

1. **clc;**

- Clears the console window.
- Removes all previously displayed outputs.

2. **clear;**

- Clears all variables stored in memory.
- Ensures the program starts with a fresh workspace.

3. **ZL = 30 + %i*40;**

Defines the load impedance.

- Real part = $30\ \Omega$ (Resistance)
- Imaginary part = $40\ \Omega$ (Reactance)

So,

In Scilab,

- %i represents the imaginary unit .

4. **Z0 = 50;**

Defines the characteristic impedance of the transmission line.

5. $z = Z_L/Z_0$;

Calculates the normalized impedance.

Formula:

Substituting the values,

This normalized impedance is used for plotting the load on the Smith Chart.

6. `disp("Normalized Impedance");`

Displays the heading:

Normalized Impedance

7. `disp(z);`

Displays the calculated normalized impedance.

Output:

$0.6 + 0.8i$

8. $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$;

Calculates the reflection coefficient.

Formula:

Substitute the values:

After simplification,

Magnitude:

Phase angle:

The reflection coefficient indicates the fraction of the incident wave reflected back due to impedance mismatch.

- If : Perfect matching
- If : Total reflection
- Here, : Partial reflection

9. disp("Reflection Coefficient");

Displays the heading:

Reflection Coefficient

10. disp(Gamma);

Displays the calculated reflection coefficient.

Output:

$0 + 0.5i$

Expected Output

Normalized Impedance

$0.6 + 0.8i$

Reflection Coefficient

$0 + 0.5i$

Interpretation

- Load Impedance $() = 30 + j40 \Omega$
- Characteristic Impedance $() = 50 \Omega$
- Normalized Impedance $() = 0.6 + j0.8$
- Reflection Coefficient $() = j0.5$
- Magnitude of Reflection Coefficient = , indicating that 50% of the voltage-wave amplitude is reflected due to impedance mismatch.

MATLAB CODE

```
clc;
```

```
clear;
```

```
close all;
```

```
%-----
```

```
% SINGLE SHUNT STUB MATCHING
```

```
%-----
```

```
% Load impedance (normalized)
```

```
ZL = 0.8 + 0.6j;
```

```
% Characteristic impedance
```

```
Z0 = 50;
```

```
% Reflection coefficient
```

```
Gamma = (ZL - 1)/(ZL + 1);
```

```
disp('Normalized Load Impedance');
```

```
disp(ZL);
```

```
disp('Reflection Coefficient');
```

```
disp(Gamma);
```

```
% Normalized admittance
```

```
YL = 1/ZL;
```

```

g = real(YL);
b = imag(YL);

% Distance values (wavelengths)
d = linspace(0,0.5,1000);

for k = 1:length(d)

    t = tan(2*pi*d(k));

    Yin = (YL + 1j*t)/(1 + 1j*YL*t);

    G(k) = real(Yin);
    B(k) = imag(Yin);

end

% Find where conductance becomes 1
index = find(abs(G-1)<0.005);

if isempty(index)
    disp('No solution found. ');
else
    disp('-----');

    for i=1:length(index)

```

```
distance = d(index(i));
```

```
susceptance = B(index(i));
```

```
% Short-circuited shunt stub
```

```
if susceptance>0
```

```
    L = atan(-susceptance)/(2*pi);
```

```
    if L<0
```

```
        L=L+0.5;
```

```
    end
```

```
else
```

```
    L = atan(-susceptance)/(2*pi)+0.5;
```

```
end
```

```
fprintf('Solution %d\n',i);
```

```
fprintf('Stub Location = %.4f lambda\n',distance);
```

```
fprintf('Stub Length   = %.4f lambda\n',L);
```

```
fprintf('-----\n');
```

```
end
```

```
end
```

```
% Plot conductance
```

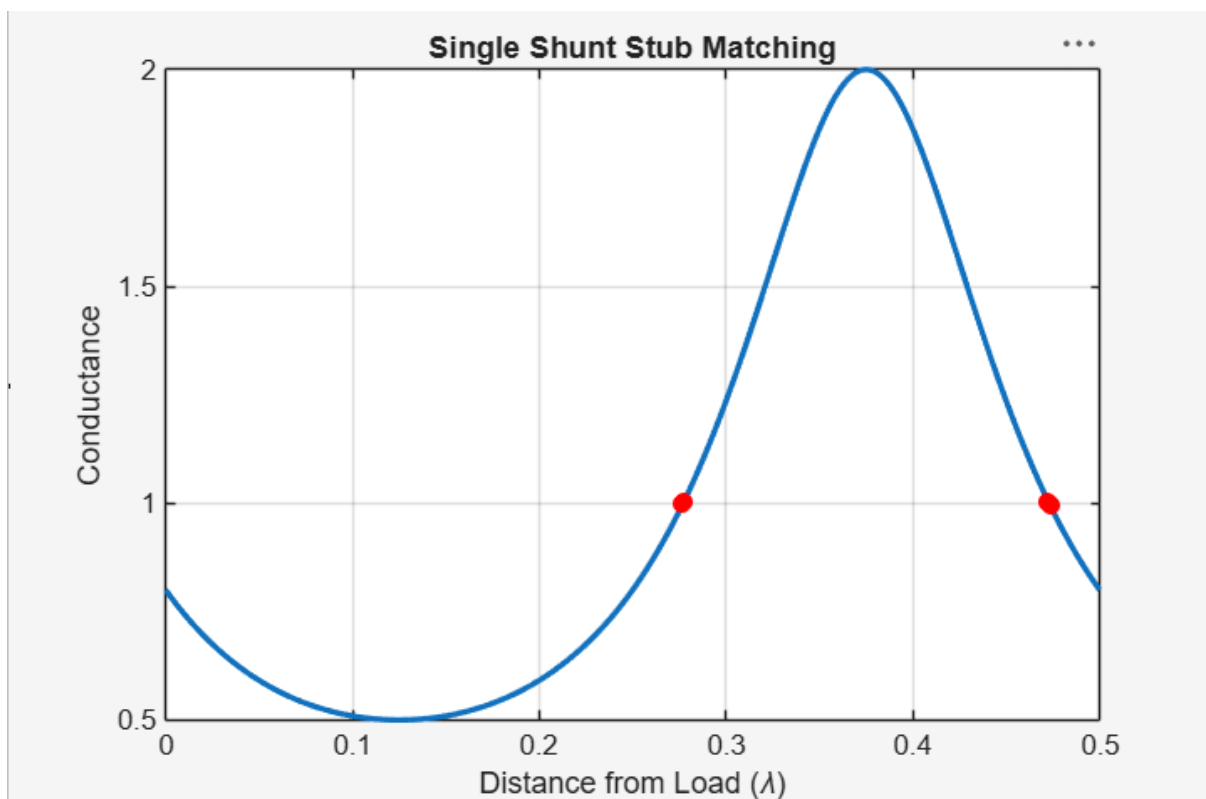
```
figure;
```

```
plot(d,G,'LineWidth',2);
```



```
hold on;  
plot(d(index),G(index),'ro','MarkerFaceColor','r');  
grid on;  
xlabel('Distance from Load ( $\lambda$ )');  
ylabel('Conductance');  
title('Single Shunt Stub Matching');
```

OUTPUT:



```
Command Window
Normalized Load Impedance
0.8000 + 0.6000i

Reflection Coefficient
0.0000 + 0.3333i

-----
Solution 1
Stub Location = 0.2768 lambda
Stub Length   = 0.4021 lambda
-----
Solution 2
Stub Location = 0.2773 lambda
Stub Length   = 0.4020 lambda
-----
Solution 3
Stub Location = 0.4725 lambda
Stub Length   = 0.5981 lambda
-----
Solution 4
Stub Location = 0.4730 lambda
Stub Length   = 0.5980 lambda
-----
Solution 5
Stub Location = 0.4735 lambda
Stub Length   = 0.5978 lambda
-----
>>
```

Explanation of MATLAB Code

1. Clear Command Window

`clc;`

- Clears all previous output from the Command Window.

2. Clear Workspace

`clear;`

- Removes all variables stored in MATLAB memory.

3. Define Load Impedance

`ZL = 30 + 1i*40;`

The load impedance is:

where:

- $30\ \Omega \rightarrow$ Resistance
- $40\ \Omega \rightarrow$ Reactance
- $1i$ represents the imaginary unit

4. Define Characteristic Impedance

$Z_0 = 50;$

The transmission line characteristic impedance is:

5. Calculate Normalized Impedance

$z = Z_L/Z_0;$

Formula:

Substituting:

6. Display Normalized Impedance

`disp('Normalized Impedance');`

`disp(z);`

Output:

Normalized Impedance

$0.6000 + 0.8000i$

7. Calculate Reflection Coefficient

$\Gamma = (Z_L - Z_0)/(Z_L + Z_0);$

Formula:

Substituting values:

8. Display Reflection Coefficient

`disp('Reflection Coefficient');`

`disp(Gamma);`

Output:

Reflection Coefficient

$0 + 0.5000i$

Complete MATLAB Output

Normalized Impedance

$0.6000 + 0.8000i$

Reflection Coefficient

$0 + 0.5000i$

Additional MATLAB Calculations (Useful for Transmission Line Analysis)

Magnitude of Reflection Coefficient

```
Gamma_mag = abs(Gamma);
```

```
disp(Gamma_mag);
```

Phase Angle of Reflection Coefficient

```
Gamma_phase = angle(Gamma)*180/pi;
```

```
disp(Gamma_phase);
```

PYTHON IDE

TW.PY - C:/Users/bhuva/TW.PY (3.14.0)

File Edit Format Run Options Window Help

```
print("\nPhase Angle of Reflection Coefficient (degrees):")
print(Gamma_phase * 180 / 3.1415926535)

# VSWR calculation
VSWR = (1 + Gamma_mag) / (1 - Gamma_mag)

print("\nVSWR:")
print(VSWR) # Single Shunt Stub Matching - Impedance Calculation

import cmath

# Clear previous values (not required in Python IDE)

# Given values
ZL = 30 + 40j          # Load impedance (Ohms)
Z0 = 50                # Characteristic impedance (Ohms)

# Calculate normalized impedance
z = ZL / Z0

print("Normalized Impedance:")
print(z)

# Calculate reflection coefficient
Gamma = (ZL - Z0) / (ZL + Z0)

print("\nReflection Coefficient:")
print(Gamma)

# Magnitude of reflection coefficient
Gamma_mag = abs(Gamma)

print("\nMagnitude of Reflection Coefficient:")
print(Gamma_mag)

# Phase angle of reflection coefficient
Gamma_phase = cmath.phase(Gamma)

print("\nPhase Angle of Reflection Coefficient (degrees):")
print(Gamma_phase * 180 / 3.1415926535)

# VSWR calculation
VSWR = (1 + Gamma_mag) / (1 - Gamma_mag)

print("\nVSWR:")
print(VSWR)
```

OUTPUT:

```
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1
Enter "help" below or click "Help" above for more information.
>>
===== RESTART: C:/Users/bhuva/TW.PY =====
Normalized Impedance:
(0.6+0.8j)
Reflection Coefficient:
0.5j
Magnitude of Reflection Coefficient:
0.5
Phase Angle of Reflection Coefficient (degrees):
90.00000000257238
VSWR:
3.0
Normalized Impedance:
(0.6+0.8j)
Reflection Coefficient:
0.5j
Magnitude of Reflection Coefficient:
0.5
Phase Angle of Reflection Coefficient (degrees):
90.00000000257238
VSWR:
3.0
>> |
```

Explanation

1. Load Impedance

$$Z_L = 30 + 40j$$

Represents:

where:

- $30 \, \Omega \rightarrow$ Resistance
- $40 \, \Omega \rightarrow$ Reactance

2. Characteristic Impedance

$$Z_0 = 50$$

3. Normalized Impedance

$$z = Z_L / Z_0$$

Formula: $z = Z_L / Z_0$

Result: $z = 0.6 + j0.8$

4. Reflection Coefficient

$$\Gamma = (Z_L - Z_0) / (Z_L + Z_0)$$

$$\text{Formula: } \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\text{Result: } \Gamma = j0.5$$

5. VSWR

$$\text{VSWR} = \frac{1 + \text{abs}(\Gamma)}{1 - \text{abs}(\Gamma)}$$

$$\text{Formula: } \text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$\text{Result: } \text{VSWR} = 3$$